THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Temperature and its Measurement** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** temperature	Temperature measures hotness.
2	**(c)** mercury	Mercury expands uniformly.
3	**(d)** conductors	Metals are good conductors.
4	**(d)** thermometer	Thermometer measures temperature.
5	**(c)** prevents mercury from falling back on its own	It keeps the reading.
6	**(d)** insulators	Wood, plastic, air.
7	**(b)** degree Celsius	Celsius scale.
8	**(c)** shaken to bring mercury below 35 °C	Reset the mercury level.
9	**(c)** aluminium	Metals conduct.
10	**(c)** it can be misleading	Past experience affects touch.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A measure of how hot or cold a body is (1); It is measured with a thermometer (1).	1+1
12	Its range is limited (1); It would break (1).	1+1
13	Metals are good conductors of heat (1); Food is cooked quickly (1).	1+1
14	It can be misleading (1); Warm hand feels lukewarm water as cold (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Wash and shake it below 35 °C (1); Place under the tongue or armpit for a minute (1); Read with the eye level to the mercury thread (1).	1+1+1
16	Cloth is a poor conductor (1); Prevents burns (1); Heat is not transferred to the hand (1).	1+1+1
17	Reduce heat exchange (1); Keeps food hot or cold (1); Helps in travel (1).	1+1+1
	OR: Keep the bulb in the liquid (1); Do not touch the vessel base (1); Read with the eye level (1).	1+1+1
18	Heat flows from hot to cold (1); Mixture has an intermediate temperature (1); Both finally have the same temperature (1).	1+1+1
	OR: Wool traps air (1); Air is an insulator (1); Stops body heat from escaping (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Glass tube with mercury bulb (1); Mercury expands on heating (1); Kink keeps the level (1); Shake before use (1); Do not hold by the bulb (1).	1+1+1+1+1
	OR: One hand in hot water, one in cold (1); Then both in lukewarm water (1); Hot hand feels cold (1); Cold hand feels warm (1); Therefore use a thermometer (1).	1+1+1+1+1
20	Take liquid in a beaker (1); Dip the bulb without touching the base (1); Wait for the reading to stay steady (1); Read with the eye level (1); Write with the unit (1).	1+1+1+1+1
	OR: Digital shows on a display (1); Infrared reads from a distance (1); No mercury risk (1); Quick reading (1); Used in schools and airports (1).	1+1+1+1+1

Mistakes to avoid:

- Using a clinical thermometer for hot water or for a room; it is only for the **human body**.
- Not shaking the clinical thermometer before use, so an old reading is mistaken for the new one.
- Letting the bulb of the laboratory thermometer touch the beaker, or reading it from a slant.
- Believing that our hands can measure temperature accurately; touch can be **misleading**.
- Writing that cold flows from ice to a hand; **heat** flows from the hotter to the colder object.

- Writing the wrong units or values: body temperature is about 37 degrees C, not 100 degrees C.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Temperature and its Measurement** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** bulb	Bulb ought to touch the body only.
2	**(c)** plastic	Plastic is an insulator.
3	**(d)** 37 °C	About 37 °C.
4	**(b)** while the bulb is in the liquid	It has no kink.
5	**(b)** a poor conductor of heat	It traps air.
6	**(c)** 0 °C	Ice melts at 0 °C.
7	**(c)** be immersed in the liquid without touching the bottom	Avoid wall or bottom.
8	**(d)** do not conduct heat easily	So that they stay cool.
9	**(d)** 100 °C	At normal pressure.
10	**(d)** level with the mercury thread	Avoid parallax.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Bulb should not touch the vessel walls (1); Read with the eye level with the mercury thread (1).	1+1
12	Plastic is a poor conductor (1); They remain cool to hold (1).	1+1
13	35 °C to 42 °C (1); It is used for body temperature (1).	1+1
14	Conductors: copper (1); Insulators: wood (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Heat flows to cooler surroundings (1); Till tea matches room temperature (1); Flow is from hot to cold (1).	1+1+1
16	Clinical: 35-42 °C, kink (1); Laboratory: -10 to 110 °C, no kink (1); Different uses (1).	1+1+1
17	Heat flows from air to ice (1); Ice absorbs heat (1); Changes to water (1).	1+1+1
	OR: Iron conducts heat quickly from the hand (1); Wood conducts slowly (1); We sense the rate of heat loss (1).	1+1+1
18	Double-walled with vacuum (1); Reduces conduction (1); Silvered walls reflect heat (1).	1+1+1
	OR: Has a sensor (1); Shows value on a display (1); Quick and safe (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Conductors pass heat easily (1); Insulators do not (1); Heat rods of various materials (1); Wax pin falls from the best conductor first (1); Classify them (1).	1+1+1+1+1
	OR: Maximum-minimum thermometer (1); Records the highest (1); Records the lowest (1); Daily readings noted (1); Used in forecasts (1).	1+1+1+1+1
20	Hot water has higher temperature (1); Heat flows to the cooler glass and air (1); Water cools (1); Surroundings warm slightly (1); Equal temperature at last (1).	1+1+1+1+1
	OR: Wooden handles (1); Plastic cover on cooker (1); Woollen clothes (1); Thermocol boxes (1); Reduces heat transfer (1).	1+1+1+1+1

Mistakes to avoid:

- Using a clinical thermometer for hot water or for a room; it is only for the **human body**.
- Not shaking the clinical thermometer before use, so an old reading is mistaken for the new one.
- Letting the bulb of the laboratory thermometer touch the beaker, or reading it from a slant.
- Believing that our hands can measure temperature accurately; touch can be **misleading**.
- Writing that cold flows from ice to a hand; **heat** flows from the hotter to the colder object.
- Writing the wrong units or values: body temperature is about 37 degrees C, not 100 degrees C.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Temperature and its Measurement** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** conduction	Heat travels along the spoon.
2	**(c)** body temperature	It is designed for human body use.
3	**(b)** hot to cold objects	Heat moves towards lower temperature.
4	**(c)** reduces heat transfer	Insulation reduces loss.
5	**(d)** 35 °C to 42 °C	It reads body temperature.
6	**(c)** they reach the same temperature	Thermal equilibrium.
7	**(b)** maximum-minimum thermometer	It records both.
8	**(d)** -10 °C to 110 °C	It is for lab use.
9	**(b)** conduction	Conduction in solids.
10	**(a)** digital infrared thermometer	It senses radiation.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Wool traps air (1); Air is a poor conductor so body heat is retained (1).	1+1
12	To stop the mercury from falling by itself (1); So that the reading can be noted after removal (1).	1+1
13	From hot to cold body (1); Until both reach the same temperature (1).	1+1
14	About 37 °C (1); It can vary slightly (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Heat one end of a metal rod (1); Pins fixed with wax along it (1); Pins fall one after another (1).	1+1+1
16	Touch is unreliable (1); A thermometer gives exact value (1); Fever can be known (1).	1+1+1
17	Illness (1); Cooking (1); Weather forecast (1).	1+1+1
	OR: Conductors: iron, copper, aluminium (1); Insulators: wood, plastic, rubber (1); Use as handles (1).	1+1+1
18	Body temperature above 37 °C (1); Measured by clinical thermometer (1); Doctor advised (1).	1+1+1
	OR: Expands evenly (1); Shiny and visible (1); Does not stick to glass (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) Plastic is an insulator (1); (b) Wool traps air (1); (c) Metal conducts (1); (d) Blanket insulates (1); (e) Flask reduces heat transfer (1).	1+1+1+1+1
	OR: Heat flows through solids (1); Particles pass on heat (1); Spoon end in tea heated (1); The other end becomes hot (1); No movement of matter (1).	1+1+1+1+1
20	0 °C is freezing (1); 100 °C is boiling (1); 100 equal divisions (1); Body temperature is 37 °C (1); Room temperature about 25 °C (1).	1+1+1+1+1
	OR: Clean before use (1); Shake down (1); Hold by stem (1); Wash with antiseptic (1); Store in a case (1).	1+1+1+1+1

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